

MEHMET BAHA DURSUN

Robotics Engineer | baha.dursun123@gmail.com | Odense, Denmark | D-type visa, Work permit 90 hours
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PROFESSIONAL SUMMARY

Robotics Engineer specializing in localization, SLAM, and autonomous navigation with 200+ robots deployed in real-world applications. Leading localization systems at Saha Robotik. MSc candidate in Advanced Robotics at the University of Southern Denmark. Proven experience in ROS 2, autonomous mobile robots, and team leadership. Successfully managed Turkey's largest autonomous navigation project (187,000 m²).

PROFESSIONAL EXPERIENCE

Robotics Engineer | Saha Robotik, Istanbul, Turkey | November 2023 - Present

- Lead localization and mapping system for 200+ autonomous waiter/cargo robots operating 24/7 in commercial environments.
- Managed complete project lifecycle for Turkey's first 24/7 autonomous airport guide robot at Istanbul Airport, covering 187,000 m² with 25 km minimum daily travel.
- Designed and implemented SLAM algorithms and control systems for autonomous navigation in dynamic environments.
- Led ROS training program, providing remote and on-site instruction to university students in Azerbaijan.
- Technologies: ROS 2, C/C++, Python, SLAM, Control Systems, Autonomous Navigation.

Student Researcher (Part-Time) | University of Southern Denmark (SDU), dVRK Surgical Robotics | October 2025 - May 2026

- Built the full learning-based control pipeline for a bimanual suturing task on the physical da Vinci Research Kit (dVRK), end to end: synchronized data collection, LeRobot dataset conversion, diffusion-policy training, ROS 1 deployment, and closed-loop evaluation on the real robot.
- Diagnosed a vision-encoder failure with representation probes (decoder feature-inversion adapted from LeWorldModel, Yann LeCun et al.) and improved policy behavior by moving from a ResNet to a fine-tuned DINOv2 encoder with state-dropout.
- Bridged the gap between simulated/theoretical models and real-world execution on a surgical robotics platform.
- Supervised by Associate Professor Iñigo Iturrate.
- Technologies: dVRK, ROS 1, PyTorch, Diffusion Policy, DINOv2, Python, C++.

TECHNICAL SKILLS

Programming: C/C++, Python, MATLAB | **Robotics:** ROS 2, ROS, SLAM, Localization, Sensor Fusion, Control Systems, Path Planning, Motion Planning | **Computer Vision:** OpenCV, YOLO, Object Detection | **Robot Learning:** PyTorch, Diffusion Policy, Imitation Learning | **Tools:** GitHub, Git, Linux, Docker | **Soft Skills:** Team Leadership, Project Management

EDUCATION

Master of Science in Advanced Robotics | University of Southern Denmark, Odense | 2025 - Present

- Full scholarship recipient.

Bachelor of Science in Mechatronics Engineering | Kocaeli University, Turkey | August 2019 - October 2024

- GPA: 3.2 / 4.0.

CERTIFICATIONS

IELTS — Overall Band Score: 6.5 (Listening: 7.5, Reading: 7.0, Writing: 5.5, Speaking: 5.5).

VOLUNTEER EXPERIENCE

Team Lead → Mentor | KOU-Mekatronom Autonomous Vehicle Team, Kocaeli University | November 2021 - Present

- Led a 20-member student team developing autonomous vehicle systems for international competitions.
- Selected as the only Turkish finalist in the Bosch Future Mobility Challenge (Top 24 of 160 global teams).
- Achieved 6th place in two consecutive Teknofest Robotaxi competitions.
- Managed project planning, resource allocation, and technical development for the autonomous-driving stack.
- Open-sourced project code and tutorials to support future cohorts.
- Led the team through 2024 and continue to mentor since then; as of 2026 still a mentor, with limited time on critical points.
- Technologies: C++, Python, MATLAB, OpenCV, YOLO, ROS, Computer Vision, Path Planning.

Volunteer Teacher | Bambu Voluntary Teaching Platform | January 2021 - August 2021

- Provided STEM tutoring for middle- and high-school students during the COVID-19 pandemic.

REFERENCES

Professor Metin Aydın | Kocaeli University

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Jacob Pilegaard | R&D Software Manager, Universal Robots

[Contact information to be provided upon request]

Associate Professor Iñigo Iturrate | University of Southern Denmark (dVRK project supervisor)

Profile: portal.findresearcher.sdu.dk/en/persons/iñigo-iturrate